We calculate $[\boldsymbol{f},[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]]]$ and project onto the space spanned by the first three components to find

$$\mathbf{u}_3 = \sigma\left(\frac{2\gamma p^3 - 4p^3\xi_2 + 4p^2 q + \xi_1^2 p\xi_2 + \gamma\xi_1^2 p - 2\xi_1^2 q}{D^2}\right)\partial_q$$

$$-\sigma\xi_1^2\left(\frac{\eta}{D^3}\right)\partial_p + 2\sigma\xi_1 p\left(\frac{\eta}{D^3}\right)\partial_{\xi_1}. \tag{8.54}$$

with

$$\begin{aligned}\eta = {} & 4p^5\gamma^2 - 8p^5\xi_1^2 - 4\xi_1^2 p^3 + 4p^3\gamma^2\xi_1^2 - 4p^3\xi_1^4 \\ & - 2\gamma\xi_1^2 p^2 q + 4p^2 q\xi_2\xi_1^2 - 4\xi_1^2 q^2 p - 2\xi_1^4 p + p\gamma^2\xi_1^4 - q\gamma\xi_1^4 + \xi_1^4\xi_2 q.\end{aligned} \tag{8.55}$$

Along $p = 0$, $\mathbf{u}_3$ becomes

$$\mathbf{u}_3|_{p=0} = -2\sigma q\partial_q - \sigma\left(\frac{-q\gamma\xi_1^4 + \xi_1^4\xi_2 q}{\xi_1^4}\right)\partial_p$$

$$= -2\sigma q\partial_q - \sigma q(-\gamma + \xi_2)\partial_p. \tag{8.56}$$

Together with $\mathbf{u}_0$ we obtain a linearly independent set as long as $q \neq 0$.

If $p = 0$ and $q = 0$ then we are on $\mathcal{M}_0$. If $p \neq 0$ then $\boldsymbol{u}_2$ and $\boldsymbol{u}_0$ are linearly independent. If $q \neq 0$ then $\boldsymbol{u}_3$ and $\boldsymbol{u}_0$ are linearly independent. From these we establish that the Hörmander condition holds everywhere except on a space of co-dimension 2 (that is except on the one-dimensional set $\mathcal{M}_0$).

In this case, it is possible to complete the proof by introducing a Lyapunov function ϕ to demonstrate the necessary contractivity in the ξ_2 direction. The generator of this system is

$$\mathcal{L}_{\text{isok}} = p\frac{\partial}{\partial q} + \left[F - (p^2/\Lambda)F\right]\frac{\partial}{\partial p} - \frac{p\xi_1 F}{\Lambda}\frac{\partial}{\partial\xi_1} + \mathcal{L}_{\text{N}} + \mathcal{L}_{\text{O}}$$

with $\Lambda = p^2 + \xi_1^2/2$,

$$\mathcal{L}_{\text{N}} = \frac{1}{2}\frac{\xi_1^2\xi_2 p}{\Lambda}\frac{\partial}{\partial p} + \frac{1}{2}\frac{\xi_1^3\xi_2}{\Lambda}\frac{\partial}{\partial\xi_1} - \xi_2\xi_1\frac{\partial}{\partial\xi_1} + [\xi_1^2 - \beta^{-1}]\frac{\partial}{\partial\xi_2},$$

and

$$\mathcal{L}_{\text{O}} = -\gamma\xi_2\frac{\partial}{\partial\xi_2} + \frac{\sigma^2}{2}\frac{\partial^2}{\partial\xi_2^2}.$$

Let us apply the operator $\mathcal{L}_{\text{isok}}$ to $\phi \stackrel{\text{def}}{=} \xi_2^2/2$. As this is only a function of ξ_2, many terms vanish and we are left with

$$\mathcal{L}_{\text{isok}}\phi = \mathcal{L}_{\text{N}}\phi + \mathcal{L}_{\text{O}}\phi,$$
$$\mathcal{L}_{\text{N}}\phi = [\xi_1^2 - \beta^{-1}]\xi_2,$$

and

$$\mathcal{L}_{\text{O}}\phi = -\gamma\xi_2^2 + \frac{\sigma^2}{2}.$$

Because ξ_1 is uniformly bounded due to the isokinetic constraint, we have

$$\mathcal{L}_{\text{isok}}\phi \leq -\gamma\xi_2^2 + K\xi_2 + \frac{\sigma^2}{2}.$$

For $|\xi_2|$ sufficiently large this term is negative. This contractivity, together with assumptions of boundedness of configurations (the use of periodic boundary conditions) and the isokinetic constraints on p and ξ_1, as well as the existence of a candidate measure which is positive on open sets, will ultimately ensure the ergodicity of the system, relying on arguments similar to those in [256, 257].

8.6.3 *Multiple Timestepping Using the Stochastic Isokinetic Formulation*

The numerical integration of the equations of the stochastic isokinetic system can be performed using splitting, in a style that should by now be familiar, with the proviso that the isokinetic condition itself should be satisfied exactly (i.e. to rounding error).

Molecular sampling using Langevin dynamics and other methods involves integration over the infinite cylinder $\mathcal{T}^{\text{N}_c} \times \mathbb{R}^{\text{N}_c}$. The power of the isokinetic method is that it allows us to reformulate the configurational sampling problem as integration on a compact manifold. The fact that we also slightly extend the domain by introducing auxiliary variables is more than offset by the stabilizing effect of the isokinetic constraints. One approach would be to introduce an isokinetic constraint on the entire system, but in practice it is found that no great crime is committed in configurational sampling by constraining each degree of freedom (together with associated auxiliary variables).

To develop a multiple timestepping method, we begin by supposing that the potential energy function and therefore the force can be decomposed into fast and slow terms, thus $U(\boldsymbol{q}) = U_{\text{S}}(\boldsymbol{q}) + U_{\text{F}}(\boldsymbol{q})$ and the force acting on the jth degree of freedom is $F_j = F_j^{\text{S}} + F_j^{\text{F}}$. Because the SIN(R) method is based on a non-Hamiltonian formulation, we do not obtain a symplectic integrator (in the absence